

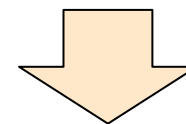
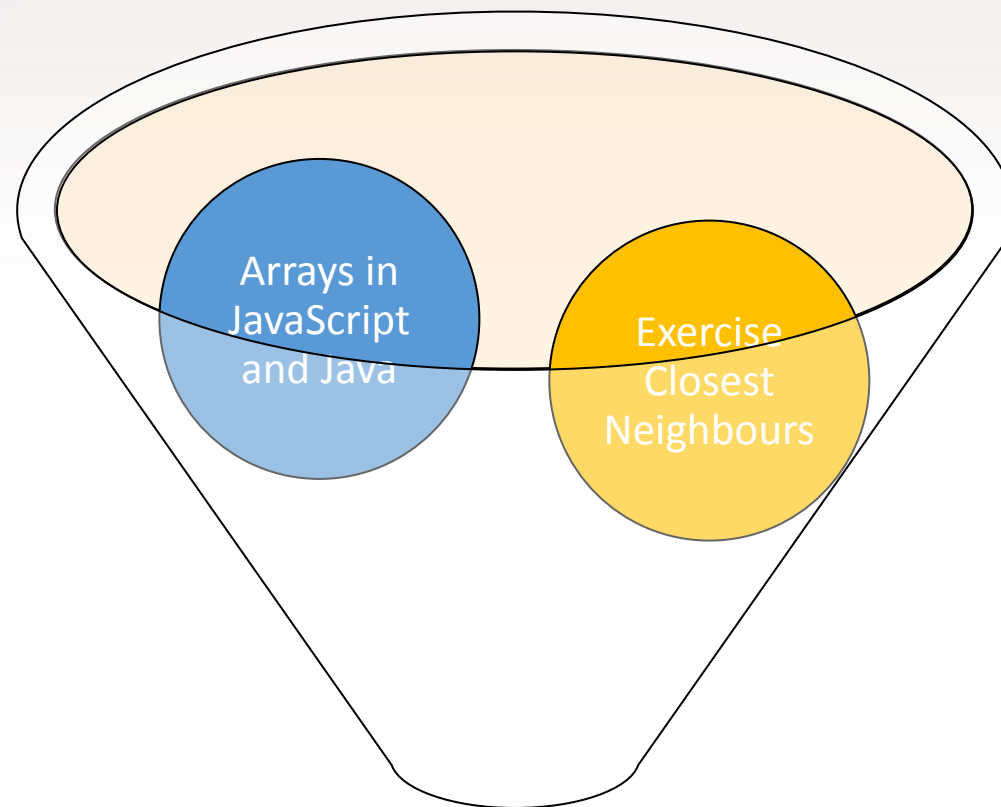
Summarizer

24 - 01 - 2023

Ricardo Ortins



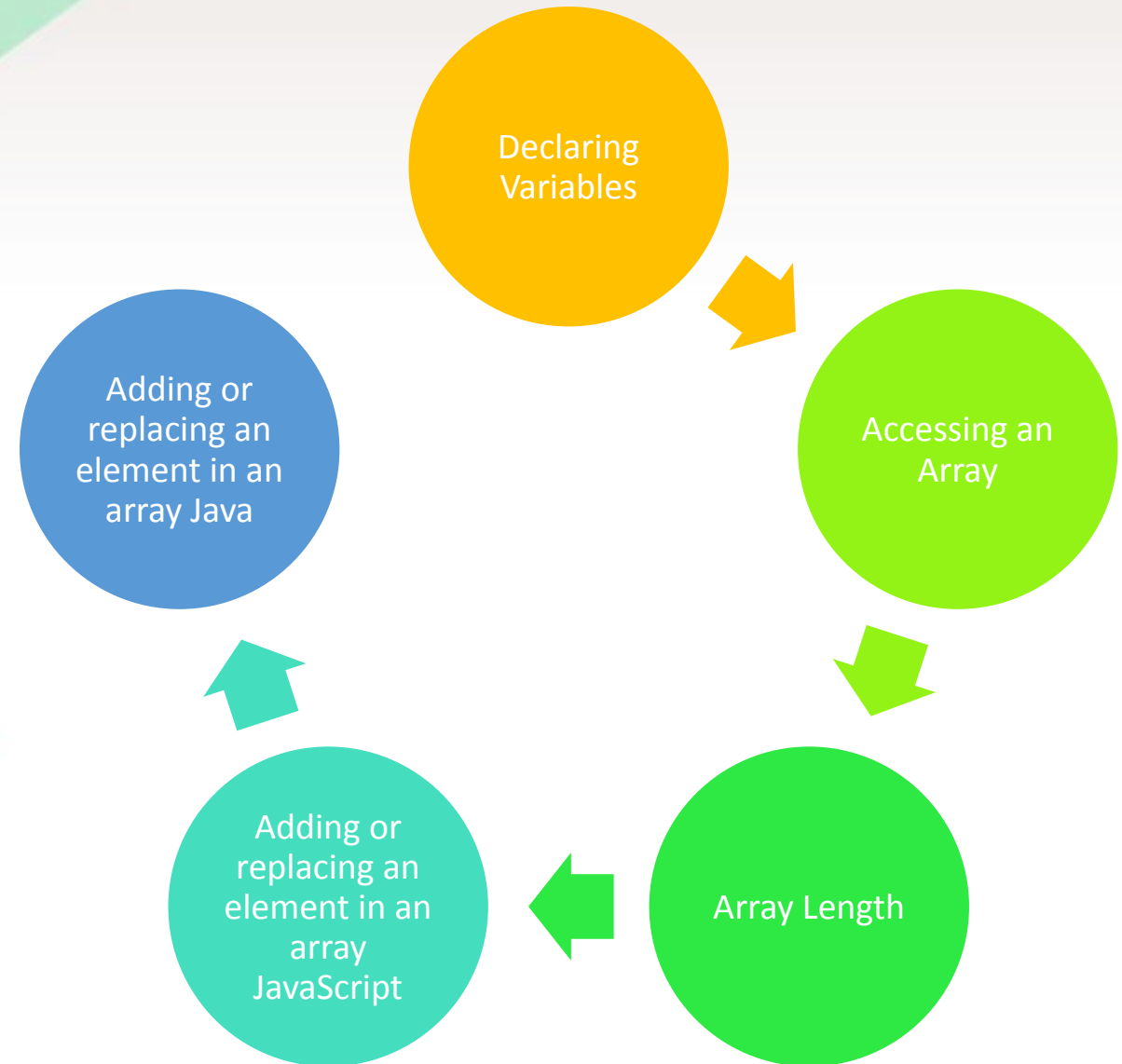
ACADEMIA DE CÓDIGO



Summarizer



Arrays In Java And JavaScript



Declaring
Variables

Accessing an
Array

Array Length

Adding or
replacing
JavaScript
(Array)

Adding or
replacing Java
(Array)

JavaScript:

```
var heroes = ["Batman", "Superman", "Spider-man"];
```

```
var numbers = [1, 2, 3, 4, 5];
```

```
var Boolean = [true, false, true];
```

Java:

```
String[] heroes = {"Batman", "Superman", "Spider-man"}; or String[] contacts = new String[4];
```

```
Int[] numbers = {1, 2, 3, 4, 5};
```

```
Boolean[] boolArray = {true, false, true};
```



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Declaring
Variables

Accessing an
Array

Array Length

Adding or
replacing
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(Array)

Adding or
replacing Java
(Array)

JavaScript:

```
var heroes = ["Batman", "Superman", "Spider-man"];
```

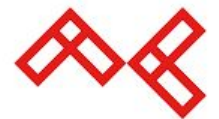
```
console.log(heroes[0]);
```

 output: **Batman**

Java:

```
String[] heroes = {"Batman", "Superman", "Spider-man"};
```

```
System.out.println(heroes[0]);
```

 output: **Batman**

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Declaring
Variables

Accessing an
Array

Array Length

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(Array)

Adding or
replacing Java
(Array)

JavaScript:

```
var heroes = ["Batman", "Superman", "Spider-man"];
```

```
console.log(heroes.length);
```

 output: 3

Java:

```
String[] heroes = {"Batman", "Superman", "Spider-man"};
```

```
System.out.println(heroes.length);
```

 output: 3

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Declaring
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```
// Add to the start of an array  
Array.unshift(element);
```

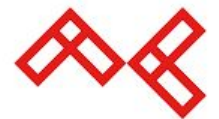
```
// Add to the end of an array  
Array.push(element);
```

```
// Add to a specified location  
Array.splice(start, 0, "element");
```

```
var heroes = ["Batman", "Superman", "Spider-man"];
```

```
heroes.splice(1, 0, "Hulk");
```

```
console.log(heroes); // ["Batman", "Hulk", "Superman", "Spider-man"];
```



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Declaring
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Array Length

Adding or
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(Array)

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(Array)

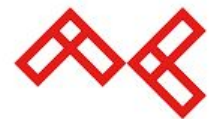
Add



Remove

Elements

Replace



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Declaring
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Accessing an
Array

Array Length

Adding or
replacing
JavaScript
(Array)

Adding or
replacing Java
(Array)

```
public class MyClass {  
    public static void main(String args[]) {
```

```
        int arr[] = {1,2,3,4};
```

```
        System.out.println("Before update " + arr[2]);
```



output: 3

```
        arr[2] = 9;    //updating the value
```

```
        System.out.println("After update " + arr[2]);
```



output: 9

```
    }  
}
```



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```
panic(cpu 1 caller 0xfffff7f9b408ca5): "DSMOS: SMC returned incorrect key: 98a6461a80ffffff2976291a80ffffff8021572c80f
5f2c80ffffff0500000000000000500000000500000050be9d2291ffffff0028602c80ffffff"@/BuildRoot/Library/Caches/com.apple.xbs/S
ntStealMacOS/DontStealMacOS-26/Dont_Steal_MacOS.cpp:219
Debugger called: <panic>
Backtrace (CPU 1), Frame : Return Address
0xfffff91229dbc60 : 0xfffff8019cf210c
0xfffff91229dbce0 : 0xfffff7f9b408ca5
0xfffff91229dbe20 : 0xfffff7f9b408b26
0xfffff91229dbe30 : 0xfffff801a29746e
0xfffff91229dbe80 : 0xfffff801a29895a
0xfffff91229dbf10 : 0xfffff801a2987c6
0xfffff91229dbf60 : 0xfffff801a29a1de
0xfffff91229dbfb0 : 0xfffff8019ca0af7
    Kernel Extensions in backtrace:
        com.apple.Dont_Steal_Mac_OS_X(7.0)[B97F871A-44FD-3EA4-BC46-8FD682118C79]@0xfffff7f9b408000->0xfffff7f9b40cf
        dependency: com.apple.kec.corecrypto(1.0)[25C67059-A2B6-328F-BCCF-F81CC7FFC10F]@0xfffff7f9a874000
        dependency: com.apple.driver.AppleSMC(3.1.9)[969D80B2-E714-3145-95B0-F61627E0EE4D]@0xfffff7f9b05b000

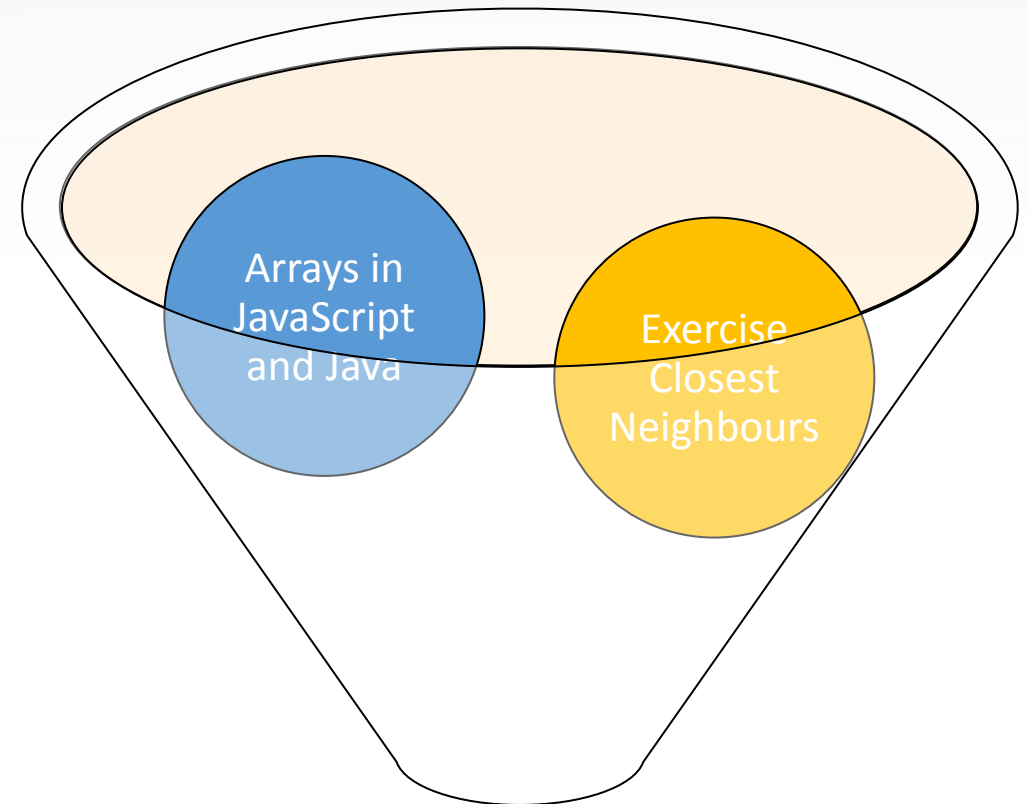
BSD process name corresponding to current thread: kernel_task

Mac OS version:
16D32

Kernel version:
Darwin Kernel Version 16.4.0: Thu Dec 22 22:53:21 PST 2016; root:xnu-3789.41.3~3/RELEASE_X86_64
Kernel UUID: C67A8D03-DEAC-35B8-8F68-06FF7B687215
Kernel slide:      0x0000000019a00000
Kernel text base: 0xfffff8019c00000
__HIB text base: 0xfffff8019b00000
System model name: VMware7,1 (H97M-ITX/ac)

System uptime in nanoseconds: 28081021858
█
```





Summarizer



Exercise Closest Neighbours

{ Exercise }

Closest Neighbours

Find the two neighbouring numbers in an array with the smallest difference to each other

```
public static void main(String[] args) {  
    int[] myArray = {0, 5, 1209, 6, 2, 111, 112, 33};  
  
    // hint: remember Math.abs();  
    // print result here  
}
```



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J ClosestNeighbours.java

```
1  public class ClosestNeighbours {
2  public static void main(String[] args) {
3
4      int[] myArray = {0, 5, 1209, 2, 2, 111, 112, 33};
5
6      int min = 0;
7      int act = 0;
8      int[] ind = new int[2];
9
10     // hint: remember Math.abs();
11     // print result here
12
13     for (int i = 0; i < myArray.length-1; i++) {
14
15
16         act = Math.abs(myArray[i] - myArray[i+1]);
17
18
19         if (act < min || min == 0 && i == 0) {
20             min = act;
21             ind[0] = myArray[i];
22             ind[1] = myArray[i+1];
23         }
24
25     }
26     System.out.println("Min: " + min + "\nInd0: " + ind[0] + "\nInd1: " + ind[1] );
27 }
28 }
29
```



